

Expanding Graduate Computer Science Pathways in Mississippi

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Introduction

Women and many racial and ethnic groups remain underrepresented in graduate computer science (CS). In the 2023–2024 academic year, women accounted for 25.5% of enrolled CS doctoral students whose gender was reported. Among doctoral students with reported residency and race or ethnicity, Black U.S. residents accounted for 1.6% and Hispanic U.S. residents for 2.2%; international students accounted for 68.1% (Computing Research Association [CRA], 2025).

One point of entry into graduate CS study is a bridge program for students whose prior degrees are outside the field. This paper describes the University of Southern Mississippi (USM) pathway and examines its first two years of student records. The records document access, selected course outcomes, and transitions to the M.S. program, but do not yet establish degree completion or the program's long-term effect on workforce diversity.

Literature review and program rationale

Graduate participation depends on opportunities accumulated before and after admission. Undergraduate research can introduce students to faculty, technical work, and graduate expectations; institutional climate and teaching practices affect persistence (Mickelson et al., 2024; Ojha et al., 2022). Financial support can influence prospective students' willingness to enroll, although this has not been studied in the context of CS-graduate study only (Templeton & Korchagin, 2025). Mentors and peers also help students navigate research expectations after entry (Ojha et al., 2022; Stout et al., 2017). Admissions alone cannot explain every disparity: one study of graduate CS research groups found nationality-related differences in the patterns examined but no evidence of gender bias in those patterns (Kalhor et al., 2023).

Bridge programs address a particular barrier within this larger pathway: applicants from other majors may lack the computing prerequisites normally expected for a CS master's degree. USM joined the MS Pathways to Computing Consortium and developed graduate-level preparation to offer students a route toward the M.S. without a second bachelor's degree (Arnold, 2024). USM identifies CSC 500, Fundamentals of Programming, and CSC 501, Fundamentals of Computing, as bridge courses. CSC 513, Algorithms, is a required, and the final bridge course, for students following the two new bridge courses.

Preparation and persistence

The pathway to graduate CS begins before the application. Undergraduate research can provide technical practice, faculty relationships, and a clearer understanding of research careers. Yet previous research experience does not necessarily prepare students for all doctoral expectations. In interviews with underrepresented computing doctoral students, Ojha et al. (2022) found that students learned those expectations through earlier research, advisors, and peers, with mentoring continuing after admission. For students whose first degree is outside CS, foundational coursework may address technical prerequisites, while advising can help make the expectations of graduate study explicit.

The undergraduate environment also shapes who remains interested in computing. Mickelson et al. (2024) examined persistence among underserved CS undergraduates and emphasized organizational climate, curriculum, teaching practices, and interactions with faculty and peers. While this research concerns undergraduate students rather than USM bridge participants, it identifies institutional conditions worth examining for bridge students.

Mississippi-specific research further supports examining computing participation as a pathway rather than a single point of entry. Lee, Ivy, and Stamps (2019) described efforts to expand equitable access to computing education in Mississippi through connected K-12 and workforce-development opportunities, emphasizing the importance of creating multiple routes into computing. In a later longitudinal study, Jaiyeola et al. (2022) examined educational pathways associated with successful transitions into the computing workforce, with particular attention to women and Black participants. Together, these studies strengthen the rationale for following students across educational transitions and evaluating not only entry into computing programs but also persistence and progression toward computing careers. Lee, Sun, and Lynn (2019) similarly described the Mississippi Coding Academies as a statewide partnership among higher education, nonprofit organizations, K-12 schools, industry, and other stakeholders designed to expand access to computing and technical education for under-resourced residents. Their work illustrates how alternative computing pathways can connect educational access with workforce development.

Admissions and financial access

Graduate recruitment and admissions draw on preparation, research opportunities, recommendations, and knowledge of the application process. Kalhor et al. (2023), studying admissions patterns in research groups at 25 highly ranked North American CS universities, reported nationality-related differences but did not find evidence of gender bias. Their findings caution against assigning a single cause to graduate enrollment disparities. A bridge pathway changes one admissions condition by supplying a structured route to the prerequisites for advanced CS study.

Affordability can also affect whether prospective students view graduate education as possible. In a survey experiment with Black, Latinx, and Native American bachelor's degree holders, Templeton and Korchagin (2025) found that financial aid opportunity increased reported enrollment likelihood and preference for full-time study. This research did not evaluate a CS bridge program, but it supports examining how clear funding information, tuition, and time to degree interact with academic access. More recent computing-specific evidence reinforces this broader finding. Kumar et al. (2026) examined a multi-institutional graduate pathways program at three public research universities in the southeastern United States and found that participants described financial support, mentoring, advising, and peer connections to graduate education as highly important. Their findings suggest that financial access may be most effective when paired with informational and social supports.

Mentoring belonging and institutional context

Academic preparation alone may be insufficient for persistence. Ojha et al. (2022) found that advisors and peers helped doctoral computing students interpret research expectations. Stout et al. (2017) evaluated the Grad Cohort workshop for women graduate students in computing, illustrating how organized mentoring and professional networks can support identity and development beyond a student's home department. These studies present opportunities for further data collection and analysis.

Evidence from bridge programs

Research on master's pathways for non-CS graduates is directly relevant to USM's design. Brodley et al. (2020) described Northeastern University's two-semester Align bridge, which combines foundational CS courses with support activities before students join traditional

M.S. coursework. Their program report presented enrollment, demographic, academic, and employment outcomes. In a ten-year retrospective, Schmidt et al. (2025) attributed the program's sustained success to dedicated advising, retention and belonging activities, and faculty recruited for bridge students. These findings suggest that evaluation should follow both academic milestones and the support ecosystem. A second model is the Illinois Computing Accelerator for Non-specialists (iCAN), a one-year online graduate certificate for students who already hold a bachelor's degree but have little or no computing background. Williams et al. (2022) described a curriculum in programming, algorithms, and data structures, a cohort-based learning model, and a mechanism for transferring some graduate credits into later computer science graduate study. Together with the Align model, iCAN shows that institutions are using multiple structures to create graduate-level entry points for students without prior CS degrees.

Together, this literature frames the bridge program as one part of a longer educational pathway. USM's present data can describe who entered, bridge courses taken, and some transitions to the M.S. program. Longer tracking and participant experiences are needed to test whether students complete the degree and how preparation, funding, mentoring, and belonging influence their progression.

Evaluation approach

This retrospective evaluation draws on student-tracking data from 2024 to 2026. The data includes 54 distinct records from Spring 2024 through Spring 2026. Two Spring 2024 records are explicitly marked not to count because the students dropped; the demographic summary therefore uses 52 records. The records describe students and selected academic statuses.

Findings

The 52 tracking records show a range of academic backgrounds. Men accounted for 34 students (65.4%) and women for 18 (34.6%). The largest recorded race or ethnicity categories were White (22, 42.3%) and Black (17, 32.7%). Nine students (17.3%) were classified as non-resident aliens. The remaining records comprised two unknown, one Asian, and one student recorded as two or more races.

Table 1. Demographic characteristics of included bridge-program tracking records.

Characteristic	Students	Percent
Gender		
Male	34	65.4%
Female	18	34.6%
Race ethnicity or residency		
White	22	42.3%
Black	17	32.7%
Non-resident alien	9	17.3%
Unknown	2	3.8%
Asian	1	1.9%

Two or more races	1	1.9%
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Electronics and Communications Engineering was the common undergraduate major, appearing for five students. Business-related, non-CS computing majors, and engineering backgrounds also appeared frequently.

In the separately reported Fall 2024 cohort of 10 students, three had entered the M.S. program by Fall 2025, four remained in the bridge, and three were no longer enrolled in either program. These figures describe one cohort at one follow-up point; the records do not explain individual reasons for departure.

Table 2. Fall 2024 bridge cohort status by Fall 2025.

Fall 2024 cohort status by Fall 2025	Students
Entered M.S. program	3
Continued in bridge	4
No longer enrolled in either	3

The reporting materials also document passes by all students enrolled in four selected course instances: CSC 500 (3 of 3) and CSC 513 (2 of 2) in Fall 2025, and CSC 500 (5 of 5) and CSC 513 (3 of 3) reported bridge courses in Spring 2026.

Discussion and next steps

These early records show that students with varied prior degrees entered USM's graduate CS pathway and that some progressed to the M.S. program. They also show why access and completion must be measured separately. The Fall 2024 cohort includes continuing students and students no longer enrolled, while selected course counts say little about eventual degree attainment.

Future work will benefit from data that includes entry date, course history, bridge completion status, M.S. transition date, and degree outcome. A second phase could examine preparation, mentoring, belonging, graduate-school knowledge, and educational intentions through surveys or interviews, subject to appropriate Institutional Review Board review. These experiences have not been measured in the current records, so no evidence about changes in student confidence or belonging is presented. Other literature also suggests that future evaluation should document supports surrounding the academic pathway. In particular, Kumar et al. (2026) identify financial, informational, mentoring, and peer supports as relevant to graduate-pathway decisions, while Williams et al. (2022) demonstrate the value of tracking bridge participation through subsequent graduate enrollment. These measures would complement USM's existing course and transition records.

The program offers an additional route toward graduate CS for students from other fields. Its long-term contribution to degree completion and Mississippi's computing workforce remains an open question that consistent tracking and participant evidence can address. This longitudinal perspective is consistent with prior Mississippi-focused work that treats broadening participation

as a sequence of educational and workforce transitions rather than a single recruitment event (Lee et al., 2019; Jaiyeola et al., 2022)

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